

Cambridge (CIE) IGCSE Computer Science

Paper 1: Computer Systems (Set A)

Tuesday 6 May 2025

Afternoon (Time: 1 hour 45 minutes)

Total marks

/ 72

Instructions

- Try to complete this mock exam paper in one sitting, under exam conditions. Use all the time available and check your answers to each question at the end before submitting.
- Remember this is PRACTICE. Mistakes are fine and will help you improve in time for the real exam - just do your best.
- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in brackets [].
- No marks will be awarded for using brand names of software packages or hardware.
- Calculators must not be used in this paper.

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1 (a) The paragraph describes the use of output devices in a modern office.

Complete the paragraph using the most appropriate terms from the list.

Not all of the terms in the list need to be used.

- **actuator**
- **digital light processing (DLP) projector**
- **inkjet printer**
- **laser printer**
- **light emitting diode (LED) screen**
- **liquid crystal display (LCD) projector**
- **liquid crystal display (LCD) screen**
- **speaker**
- **3D printer**

In a modern office setup, various devices are used to enhance productivity and communication.

For displaying presentations and videos, a _____ (1) is commonly used as it projects high-quality images onto a large screen.

When it comes to producing physical documents, a _____ (2) printer is preferred for its fast and precise text printing, while an _____ (3) printer is chosen if budget is a concern.

For outputting clear audio during meetings or multimedia sessions, a _____ (4) ensures everyone can hear the sound clearly.

Additionally, a _____ (5) screen is often used in office monitors for its bright and sharp display, making it ideal for daily work.

.....

.....

(5 marks)

- (b) A farm has an automated drinking system for its animals. The drinking system has a water bowl that contains the water. When the water bowl is empty, it is automatically refilled.

The system uses a **sensor** and a microprocessor.

Identify the most appropriate **sensor** for this system.

(1 mark)

- (c) An office has an automated lighting system. When movement is detected in the office, the lights are switched on. If movement is not detected for a period of two minutes, the lights are switched off.

The system uses a **sensor** and a microprocessor.

Identify the most appropriate **sensor**.

(1 mark)

2 (a) A school network has several computers.

Each computer in the network has a media access control (MAC) address.

Hexadecimal is used for MAC addresses.

Part of a MAC address is given.

97-5C-E1

Each pair of digits is stored as binary in an 8-bit register.

Complete the binary register for these two pairs of digits.

97								
----	--	--	--	--	--	--	--	--

5C								
----	--	--	--	--	--	--	--	--

(4 marks)

(b) Give two other uses of hexadecimal in Computer Science.

(2 marks)

(c) Another value is stored as binary in a register.

0	1	0	1	0	0	1	0
---	---	---	---	---	---	---	---

A **logical left shift** of **two places** is performed on the binary value.

Complete the binary register to show its contents after this logical left shift.

..... (1 mark)

(d) **State one** effect this logical shift has on the binary value.

..... (1 mark)

(e) Binary is a number system used by computers.

Two 8-bit binary values are added.

The result of this calculation needs to be stored in an 8-bit register.

The denary result of this calculation is 301.

This generates an error

State the name of this type of error and **explain** why this error occurs.

Error name:	
Explanation:	

.....
.....

(3 marks)

- 3 (a)** An art gallery uses **secure socket layer (SSL)** to provide a secure connection when selling art on its website.

Describe the process of SSL and explain how it provides a secure connection.

(6 marks)

- (b)** A student is concerned about the threats to their computer when using the internet.

The student wants to use some **security solutions** to help protect the computer from the threats.

Describe how each of the following security solutions will help protect the computer.

- Anti-malware
- Two-step verification
- Passwords

(6 marks)

(c) Adele has a **software-based firewall** installed on her mobile device.

The firewall gathers data about the traffic coming into and going out of her mobile device.

Explain how the firewall uses the gathered data to keep the mobile device more secure.

(2 marks)

4 (a) A student uses the internet for their schoolwork to research what is meant by pharming.

Storing **cookies** is one function of the web browser.

Give three other functions of the web browser.

(3 marks)

(b) Explain the differences between **HTTP** and **HTTPS**.

Include an **example** scenario where HTTPS would be preferred over HTTP

(4 marks)

5 (a) A computer has a Von Neumann architecture.

State three components that are part of the central processing unit (**CPU**) in this computer.

Accumulator (ACC)	Hard disk drive (HDD)
Memory address register (MAR)	Program counter (PC)
Random access memory (RAM)	Read only memory (ROM)
sensor	Solid state drive (SSD)

(3 marks)

(b) The computer has a single core CPU.

State one purpose of a core in a CPU.

(1 mark)

(c) A student uses a laptop and a calculator for schoolwork.

The laptop has a central processing unit (CPU) that performs the fetch–decode–execute cycle.

The CPU has several components, including the memory data register (MDR) and the arithmetic logic unit (ALU).

Describe how the MDR and the ALU are used in the fetch–decode–execute cycle.

(5 marks)

(d) An office has an automated lighting system. When movement is detected in the office, the lights are switched on. If movement is not detected for a period of two minutes, the lights are switched off.

The system uses a **sensor** and a **microprocessor**.

Describe how the **sensor** and the **microprocessor** are used in this automated lighting system.

(6 marks)

(e)

Instruction	Mnemonic	Binary Code	Command
Add	ADD	10100001	Adds a value to the value currently stored in the accumulator (ACC)
Subtract	SUB	00100010	Subtracts a value from the value stored in the accumulator
Load	LDA	10111111	Loads the value stored in a memory location into the accumulator
Store	STA	01100000	Stores the value in the accumulator in a specific location in memory
Stop	HLT	00000000	Stops the program

A program consists of the following sequence of instructions, written using the mnemonics from the table:

1. **LDA**
2. **ADD**
3. **STA**
4. **HLT**

Explain what happens during the execution of this program, assuming it starts with an **initial value of 5** in the memory location loaded by the LDA instruction and the **value to add is 3**.

(4 marks)

- 6 (a)** The website hosts videos that users can stream. The company uploads new videos to the website.

The videos are compressed before they are uploaded to the website.

Tick (✓) one box to show which statement is a benefit of compressing the videos.

- A.** Data is encrypted.
- B.** Duration of each video will be reduced.
- C.** Less storage space on the web server is required.
- D.** More bandwidth is required when viewing the videos.

(1 mark)

- (b) Give two** methods of compression that could be used to compress the videos.

.....

.....

(2 marks)

- (c)** An image is stored on a computer. The image is 16-bit colour and is 100 pixels high and 150 pixels wide.

Calculate the file size of the image in bytes.

Show all your **working**.

.....

.....

.....

(3 marks)

7 (a) Identify the type of data transmission used in a USB connection.

.....
(1 mark)

(b) Julia inputs personal data into her computer.

She stores three copies of the data using a hard disk drive (HDD), a solid state drive (SSD) and a USB flash memory drive.

Julia uses a USB connection to transfer data onto her USB flash memory drive

One benefit of using a USB connection is that it is a universal connection.

State two other benefits of using a USB connection.

.....
(2 marks)

(c) A student has a sound file that is too large to be stored on their external secondary storage device. The student compresses the sound file to make the file size smaller.

The student sends the sound file to a friend. The file is transmitted across a network that uses packet switching.

Explain how the file is transmitted using packet switching.

.....
(5 marks)